Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

Miyu Feng, Shiwei Guo, Tiebin Mi, and Robert Caiming Qiu

***Abstract*—This paper investigates reference-assisted signal recovery for holographic reconfigurable intelligent surface (RIS) receivers with intensity-only measurements. A known reference wave interferes with the received field before square-law detection, thereby converting the resulting inverse problem into affine phase retrieval and removing the global-phase ambiguity. We derive an exact secant identity, continuous-channel identifiability conditions, and an energy-domain minimum-distance criterion for finite-alphabet detection. A four-phase reference construction further yields an explicit stability lower bound. Based on these results, a two-stage receiver is developed: a damped Gauss–Newton method jointly estimates the directions of arrival and complex path gains, while a reference-assisted Gerchberg–Saxton detector recovers data symbols, with maximum-likelihood detection serving as a benchmark. A Cramér–Rao lower bound characterizes the attainable angular accuracy. Simulations verify the predicted effects of reference design, training length, array size, and measurement diversity on channel estimation and symbol detection.**

***Index Terms*—Reconfigurable intelligent surface, holographic receiver, phase retrieval, signal recovery, intensity-only measurements.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) provide a programmable electromagnetic interface for controlling wireless propagation with low-complexity surface elements. Their operating principles, communication models, and potential gains in coverage and energy efficiency have been extensively studied [1]–[9]. Beyond conventional reflecting architectures, densely sampled large intelligent and holographic surfaces can act as spatially continuous apertures, offering high spatial resolution and new wave-domain processing capabilities [10]–[13]. Recent studies have further explored holography-inspired surface control, wave-domain DOA estimation, and metasurface-enabled sensing and localization [14]–[18]; wavenumber-domain processing provides a complementary framework for modeling and processing quasi-continuous holographic apertures [19].

A central implementation challenge is that acquiring the complex field at every surface element conventionally requires coherent radio-frequency chains, local-oscillator distribution, and in-phase/quadrature sampling. Power-detector-based architectures instead retain only intensity measurements and can substantially simplify the receiving hardware [20], [21]. Square-law or direct detection is well established in optical and infrared communication systems [22], [23]; when used for complex-valued radio signals, however, it discards the phase that carries essential channel and symbol information. The resulting inverse problem is therefore a form of phase retrieval.

Phase retrieval has a long history in optics and signal processing [24], [25]. Convex lifting and semidefinite relaxations provide recovery guarantees at the cost of high-dimensional optimization [26]–[28], whereas alternating-minimization and gradient-based methods offer more scalable nonconvex alternatives [29], [30]. Recent deep-unfolding approaches further combine interpretable iterative structure with learned signal priors [31]. In parallel, fundamental results have characterized the number and structure of measurements required for uniqueness and stable recovery [32]–[36]. These results clarify the generic phaseless inverse problem, but its unavoidable global-phase ambiguity remains incompatible with the direct recovery of absolute communication symbols.

A known reference wave can remove this ambiguity by interfering with the unknown field before square-law detection. This principle originates from holography, where the object and reference waves form an intensity pattern that encodes their relative phase [37], [38]; controlled reference phases can provide additional complex-field information [39]. Algebraically, the reference converts homogeneous magnitude measurements into affine magnitude measurements, connecting the receiver considered here to affine phase retrieval [40]. Unlike a conventional coherent local oscillator used after antenna combining, the reference in this architecture participates directly in the element-wise interference pattern.

Phaseless recovery has also appeared in communication problems such as OFDM channel estimation [41]. Meanwhile, RIS channel acquisition and codebook design have been studied for conventional complex-baseband observations [42]–[44]. These works establish useful estimation and training mechanisms, but they do not resolve the identifiability of finite-alphabet symbols from reference-assisted intensity measurements. In particular, generic continuous-signal guarantees do not directly describe when two constellation vectors produce identical or nearly identical energy patterns under the coupled effects of the reference wave, waveform sampling, and RIS array manifold.

This paper addresses that gap through a unified reference-assisted phase-retrieval framework for holographic RIS receivers. We first separate field-domain injectivity from reference-induced intensity separation and derive conditions under which finite-alphabet symbols are uniquely represented by the observed energies. We then introduce an energy-domain minimum-distance metric that quantifies noise robustness and

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exposes the joint roles of reference diversity and RIS sampling. Finally, we develop practical channel-estimation and symbol-detection algorithms whose performance can be interpreted through these identifiability and separation properties.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

II. System Model

We consider a holographic RIS receiver in which L single-antenna users transmit complex baseband symbols to a surface composed of multiple receiving elements. Each element senses the intensity of the interference pattern formed by the received field and a known reference wave generated by a local oscillator [21]. The receiver first estimates the complex channel parameters and directions of arrival (DOAs) and then uses the resulting channel estimate to recover the transmitted symbols.

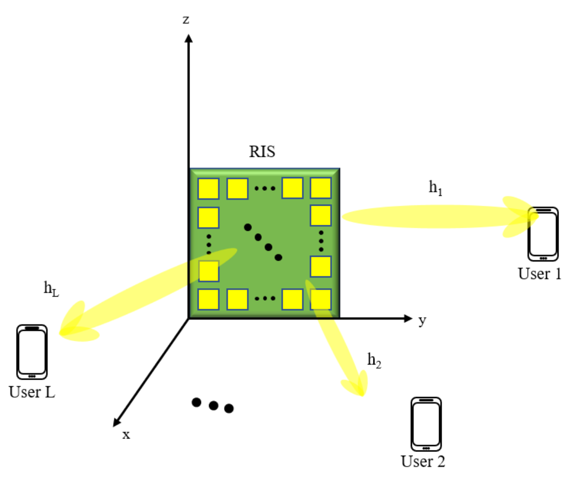

Fig. 1. System model of the holographic RIS receiver with intensity-only measurements.

A. Transmitter Model

The K symbols transmitted by user l in one signaling block are collected in $\mathbf{s}_l = [s_{l,0}, \dots, s_{l,K-1}]^T \in \mathcal{S}^K$, where $\mathcal{S} = \{c_0, \dots, c_{M_{\text{mod}}-1}\} \subset \mathbb{C}$ is an equiprobable constellation normalized to $\frac{1}{M_{\text{mod}}}\sum_{q=0}^{M_{\text{mod}}-1} |c_q|^2 = \mathbb{E}\{|s_{l,k}|^2\} = 1$. After pulse shaping, the complex baseband waveform is

$$x_l(t) = \sqrt{P_l}\left(\sum_{k=0}^{K-1} s_{l,k}p(t-kT_s) + \xi_l q(t)\right), \quad (1)$$

Here, P_l is the transmit power, $p(t)$ is the pulse-shaping filter, T_s is the symbol duration, and the optional deterministic term $\xi_l q(t)$ is set to zero when no transmit-side pilot or bias is required. The corresponding real passband waveform radiated by user l is

$$x_l^{\text{RF}}(t) = \sqrt{2}\,\text{Re}\{x_l(t)e^{j2\pi f_c t}\}, \quad (2)$$

where f_c is the carrier frequency. Under the usual carrier-period averaging, the factor $\sqrt{2}$ makes the average power of $x_l^{\text{RF}}(t)$ equal to the envelope power $|x_l(t)|^2$. Thus, $x_l(t)$ contains the pulse-shaped amplitude and phase information, while the carrier term translates that waveform to RF for propagation.

B. Far-Field Channel Model

As shown in Fig. 1, the holographic RIS is an $N_y \times N_z$ uniform planar array on the yz-plane, with $N_\text{r} = N_y N_z$ elements. The (n,m)-th element is located at $\mathbf{r}_{n,m} = [0, y_n, z_m]^T$, where $y_n = (n-(N_y+1)/2)d_y$ and $z_m = (m-(N_z+1)/2)d_z$. The centered coordinates introduce no loss of generality because a change of origin produces only a user-dependent phase that can be absorbed into the complex gain.

Under far-field propagation, user l produces a plane wave with polar angle θ_l measured from the array broadside and azimuth angle ϕ_l measured from the positive y-axis in the yz-plane. With $k_0 = 2\pi/\lambda$, the relevant tangential wavenumbers are $k_{l,y} = k_0 \sin\theta_l \cos\phi_l$ and $k_{l,z} = k_0 \sin\theta_l \sin\phi_l$. The phase at element (n,m) is

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y}y_n + k_{l,z}z_m. \quad (3)$$

The far-field amplitude is taken as constant across the aperture, while path loss, shadowing, and the user-dependent phase are absorbed into $\alpha_l \in \mathbb{C}$. Accordingly,

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l, \qquad [\mathbf{v}_l]_{\iota(n,m)} = e^{j\psi_{n,m}^{(l)}}, \quad (4)$$

where $\iota(n,m) = (n-1)N_z + m$. Collecting all users gives

$$\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] = \underbrace{[\mathbf{v}_1, \dots, \mathbf{v}_L]}_{\mathbf{V}}\underbrace{\text{diag}(\alpha_1, \dots, \alpha_L)}_{\mathbf{\Gamma}} = \mathbf{V}\mathbf{\Gamma}, \quad (5)$$

which separates the directional array response from the complex path gains.

C. Reference-Assisted Holographic Reception Model

Let $\mathbf{x}(t) = [x_1(t), \dots, x_L(t)]^T$. At RIS element i, the locally generated reference is

$$b_i^{\text{RF}}(t) = \sqrt{2}\,\text{Re}\{b_i e^{j2\pi f_c t}\}, \qquad b_i = \rho_i e^{j\chi_i}, \quad (6)$$

where ρ_i and χ_i are its prescribed amplitude and phase. Let $\mathbf{b} = [b_1, \dots, b_{N_\text{r}}]^T$ collect the reference coefficients, which are carrier aligned and phase stable over each detection interval.

Under the narrowband model, the received user field has envelope $[\mathbf{H}\mathbf{x}(t)]_i$. Its superposition with (6) gives

$$r_i^{\mathrm{RF}}(t) = \sqrt{2}\,\mathrm{Re}\big\{([\mathbf{H}\mathbf{x}(t)]_i + b_i)\, e^{j2\pi f_c t}\big\}\,. \tag{7}$$

Here, $\mathbf{H}\mathbf{x}(t)$ contains the propagation response and unknown user phases, whereas b_i is receiver controlled. The reference-assisted interference is therefore formed in the analog RF domain before square-law detection and sampling.

Each symbol is observed over M subslots of duration $\Delta_t = T_s/M$, with centers $t_{k,m} = kT_s + (m + 1/2)\Delta_t$ and intervals $\mathcal{I}_{k,m} = [kT_s + m\Delta_t, kT_s + (m+1)\Delta_t)$. Direct square-law detection and integration give

$$\begin{aligned} z_i[k,m] &= \frac{1}{\Delta_t}\int_{\mathcal{I}_{k,m}} \left[r_i^{\mathrm{RF}}(\tau)\right]^2 d\tau + w_i[k,m] \\ &\simeq \frac{1}{\Delta_t}\int_{\mathcal{I}_{k,m}} \left|[\mathbf{H}\mathbf{x}(\tau)]_i + b_i\right|^2 d\tau + w_i[k,m]. \end{aligned} \tag{8}$$

Here, $w_i[k,m]$ models the effective post-detection noise. For $\Delta_t f_c \gg 1$, the integration suppresses the $2f_c$ component, which gives the equivalent complex-envelope expression on the second line. This is an analytical representation of the measured RF energy, not coherent downconversion or I/Q sampling. If the envelope is approximately constant within a subslot, stacking the N_r element outputs yields

$$\mathbf{z}[k,m] = \left|\mathbf{H}\mathbf{x}[k,m] + \mathbf{b}\right|^2 + \mathbf{w}[k,m] \in \mathbb{R}^{N_\mathrm{r}}, \tag{9}$$

where $\mathbf{x}[k,m] = \mathbf{x}(t_{k,m})$ and the squared modulus is element-wise. The MN_r samples associated with each symbol are retained as a vector because their spatial and temporal variations provide the diversity needed for recovery.

To relate these energy samples to the transmitted symbols, define $\mathbf{D}_P = \mathrm{diag}(\sqrt{P_1}, \ldots, \sqrt{P_L})$, $[\mathbf{P}]_{kM+m,j} = p(t_{k,m} - jT_s)$, and let $\mathbf{q}$ collect the corresponding samples of $q(t)$. Stack $\mathbf{s} = [\mathbf{s}_1^T, \ldots, \mathbf{s}_L^T]^T$ and set $\boldsymbol{\xi} = [\xi_1, \ldots, \xi_L]^T$. The matrix $\mathbf{P}_{\mathrm{tx}} \triangleq \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LKM\times LK}$ maps $\mathbf{s}$ to the sampled user envelopes, while $\mathbf{x}_0 \triangleq (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q}$ represents the deterministic transmit-side component; here, $\otimes$ denotes the Kronecker product.

For general pulse shaping, neighboring symbols may overlap, so the KMN_r measurements must be processed jointly. Let $\boldsymbol{\Pi}$ reorder the user-major envelope $\mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0$ into time-major form. Stacking (9) over all subslots then yields the compact block-level model

$$\mathbf{z} = \left|\mathbf{G}\mathbf{s} + \mathbf{b}_{\mathrm{eq}}\right|^2 + \mathbf{w}, \tag{10}$$

where $\mathbf{G} \triangleq (\mathbf{I}_{KM} \otimes \mathbf{H})\boldsymbol{\Pi}\mathbf{P}_{\mathrm{tx}} \in \mathbb{C}^{KMN_\mathrm{r}\times LK}$ and $\mathbf{b}_{\mathrm{eq}} \triangleq \mathbf{1}_{KM} \otimes \mathbf{b} + (\mathbf{I}_{KM} \otimes \mathbf{H})\boldsymbol{\Pi}\mathbf{x}_0$. Thus, $\mathbf{b}_{\mathrm{eq}}$ combines the receiver reference with the deterministic transmit-side contribution, while $\mathbf{s} \in \mathcal{S}^{LK}$ and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^{KMN_\mathrm{r}}$. When intersymbol overlap is negligible, (10) separates into symbol-wise models; otherwise, it describes joint block detection.

D. Two-Phase Transmission Protocol and Unified Observation Model

The receiver uses a training phase to estimate the channel from known pilots and a data phase to detect finite-alphabet symbols using the estimated channel. Both phases follow (10), but their unknowns have different structures.

1) Training Phase: Let $\mathbf{x}_\mathrm{p}[t] \in \mathbb{C}^L$ be the known waveform in pilot subslot $t = 0, \ldots, T_\mathrm{p} - 1$, and define $\mathbf{h} = \mathrm{vec}(\mathbf{H})$. Stacking the element-wise observations and using $\mathbf{H}\mathbf{x}_\mathrm{p}[t] = (\mathbf{x}_\mathrm{p}^T[t] \otimes \mathbf{I}_{N_\mathrm{r}})\mathbf{h}$ gives

$$\mathbf{z}_\mathrm{p} = \left|\mathbf{G}_\mathrm{p}\mathbf{h} + \mathbf{b}_\mathrm{p}\right|^2 + \mathbf{w}_\mathrm{p}, \tag{11}$$

where $\mathbf{G}_\mathrm{p}$ stacks $\mathbf{x}_\mathrm{p}^T[t] \otimes \mathbf{I}_{N_\mathrm{r}}$, and $\mathbf{b}_\mathrm{p}$ and $\mathbf{w}_\mathrm{p}$ stack the reference and noise. The far-field structure in (5) will later reduce the unknown from the full vector $\mathbf{h}$ to the DOAs and complex path gains.

2) Data Transmission Phase: For a data block of K_d symbols per user, let $\mathbf{s}_\mathrm{d} \in \mathcal{S}^{LK_\mathrm{d}}$ denote the stacked symbol vector. Replacing $\mathbf{H}$ in (10) by its estimate $\widehat{\mathbf{H}}$ yields

$$\mathbf{z}_\mathrm{d} = \left|\mathbf{G}_\mathrm{d}(\hat{\mathbf{H}})\mathbf{s}_\mathrm{d} + \mathbf{b}_\mathrm{d}\right|^2 + \mathbf{w}_\mathrm{d}, \tag{12}$$

where $Q_\mathrm{d} = N_\mathrm{r}MK_\mathrm{d}$ and the known offset $\mathbf{b}_\mathrm{d}$ includes the receiver reference and any deterministic transmit-side component. Thus, (11) and (12) have the same affine intensity structure, while the former contains continuous channel parameters and the latter a finite-alphabet vector.

III. Uniqueness and Stability of Reference-Assisted Energy Detection

Reliable recovery from intensity-only measurements requires more than the existence of a solution: distinct channels or symbol vectors must produce distinct energy observations, and small measurement perturbations must not cause arbitrarily large reconstruction errors. These requirements are captured by uniqueness and stability, respectively, and are fundamental to determining whether the reference wave and RIS sampling provide sufficient information. Exploiting the quadratic structure of the reference-assisted energy mapping, we establish an exact secant-based uniqueness condition and derive lower and upper Lipschitz bounds that quantify robustness. The resulting framework applies to both continuous channel estimation and finite-alphabet symbol detection.

A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \qquad \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq \left|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\right|^2, \tag{13}$$

where Q is the total number of scalar intensity observations, D is the complex dimension of the unknown vector, $\mathbf{G} \in \mathbb{C}^{Q\times D}$, $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$, $\mathbf{b} \in \mathbb{C}^Q$, and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^Q$. Here, $\mathcal{X}$ denotes the feasible set of the unknown parameter, and the modulus and square are applied element-wise. This is an affine phase-retrieval model [40]. For full-channel estimation in the training phase, $\boldsymbol{\theta} = \mathbf{h} = \mathrm{vec}(\mathbf{H})$, $Q = N_\mathrm{r}T_\mathrm{p}$, and $D = N_\mathrm{r}L$. If the channel is represented by a lower-dimensional linear complex parameter, D is replaced by the dimension of that parameter. In the data phase, $\boldsymbol{\theta} = \mathbf{s}_\mathrm{d} \in \mathcal{S}^{LK_\mathrm{d}}$, $Q = Q_\mathrm{d}$, and $D = LK_\mathrm{d}$.

The reference changes the ambiguity class fundamentally. If $\mathbf{b} = 0$, $\mathcal{T}_{\mathbf{G},\mathbf{0}}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},\mathbf{0}}(\boldsymbol{\theta})$, so uniqueness is defined only modulo a global phase. If $\mathbf{b} \neq 0$, the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2\,\mathrm{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \tag{14}$$

contains phase-sensitive interference, so absolute uniqueness on $\mathcal{X}$ becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

For $\boldsymbol{\theta} \in \mathbb{C}^D$, define the realification $\boldsymbol{\theta}_{\mathbb{R}} = [\mathrm{Re}\{\boldsymbol{\theta}\}^T, \mathrm{Im}\{\boldsymbol{\theta}\}^T]^T \in \mathbb{R}^{2D}$. For two distinct parameters, define the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\|_2 . \tag{15}$$

The optimal lower and upper stability bounds on $\mathcal{X}$ are

$$\alpha_{\mathcal{X}} \triangleq \inf_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2}, \tag{16}$$

$$\beta_{\mathcal{X}} \triangleq \sup_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2}. \tag{17}$$

Thus, the inverse problem is globally stable on $\mathcal{X}$ if $0 < \alpha_{\mathcal{X}} \leq \beta_{\mathcal{X}} < \infty$. The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (13) gives an exact identity, which is stronger than a first-order approximation. For a point $\boldsymbol{\vartheta} \in \mathbb{C}^D$, define the real Jacobian

$$\begin{aligned}\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2\big[\, & \mathrm{Re}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta}+\mathbf{b})^*]\mathbf{G}\},\\ & -\mathrm{Im}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta}+\mathbf{b})^*]\mathbf{G}\}\big] \in \mathbb{R}^{Q\times 2D}.\end{aligned} \tag{18}$$

Lemma 1 (Exact midpoint factorization)**.** *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$, let $\boldsymbol{\mu} = (\boldsymbol{\theta}_1+\boldsymbol{\theta}_2)/2$ and $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_{\mathbb{R}}. \tag{19}$$

Proof. Applying $|a|^2 - |c|^2 = \mathrm{Re}\{(a+c)^*(a-c)\}$ element-wise gives $\mathrm{Re}\{[\mathbf{G}(\boldsymbol{\theta}_1+\boldsymbol{\theta}_2)+2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2)\}$, which equals (19) after realification. □

Theorem 1 (Pairwise uniqueness criterion)**.** *The mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is injective on $\mathcal{X}$ if and only if, for every distinct $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$,*

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1+\boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2)_{\mathbb{R}} \neq \mathbf{0}. \tag{20}$$

A necessary condition, independent of the reference, is

$$\begin{aligned}&\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing,\\ &\mathcal{D}_{\mathcal{X}} \triangleq \{\boldsymbol{\theta}_1-\boldsymbol{\theta}_2 : \boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X},\, \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2\}.\end{aligned} \tag{21}$$

Proof. The first claim follows directly from Lemma 1. For the second, if $\mathbf{G}(\boldsymbol{\theta}_1-\boldsymbol{\theta}_2) = 0$, then the two complex fields, and hence their intensities for every $\mathbf{b}$, are identical. □

Theorem 1 makes clear why $\mathrm{rank}(\mathbf{G}) = D$ is necessary for a full-dimensional continuous parameter set but is not sufficient for phase retrieval: even when $\mathbf{G}\boldsymbol{\delta} \neq 0$, its real projection determined jointly by the midpoint field and the reference may vanish. For a finite constellation, full column rank is not necessary; only the finite difference set in (21) must avoid the null space.

Let $\mathcal{M}_{\mathcal{X}} = \{(\boldsymbol{\theta}_1+\boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1,\boldsymbol{\theta}_2 \in \mathcal{X}\}$ be the midpoint set. The exact identity yields computable two-sided bounds.

Proposition 1 (Jacobian stability bounds)**.** *Define*

$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu}\in\mathcal{M}_{\mathcal{X}}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \qquad \overline{\sigma}_J \triangleq \sup_{\boldsymbol{\mu}\in\mathcal{M}_{\mathcal{X}}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \tag{22}$$

Then

$$\underline{\sigma}_J \|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2 \leq d_E(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2) \leq \overline{\sigma}_J \|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\|_2 \tag{23}$$

for all pairs in $\mathcal{X}$. Consequently, $\alpha_{\mathcal{X}} \geq \underline{\sigma}_J$ and $\beta_{\mathcal{X}} \leq \overline{\sigma}_J$.

Proof. Apply the singular-value inequalities to (19) over the midpoint set. □

The bound $\underline{\sigma}_J > 0$ is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (16), or equivalently the restricted secants in (19), is the sharp criterion. Notice also that injectivity alone need not give a useful uniform lower bound on a continuous set; this is precisely the distinction between uniqueness and stability emphasized in [36].

C. Continuous Channel Identifiability

For a continuous channel parameter, local identifiability at $\boldsymbol{\theta}$ requires

$$\mathrm{rank}(\mathbf{J}(\boldsymbol{\theta})) = 2D, \qquad Q \geq 2D. \tag{24}$$

The smallest singular value determines the local inverse sensitivity:

$$\liminf_{\|\boldsymbol{\delta}\|_2\to 0} \frac{d_E(\boldsymbol{\theta}+\boldsymbol{\delta},\boldsymbol{\theta})}{\|\boldsymbol{\delta}\|_2} = \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta})). \tag{25}$$

Local full rank does not rule out a distant ambiguous channel. Global uniqueness still requires (20) for all pairs, while a positive $\alpha_{\mathcal{X}}$ rules out both exact and near-ambiguities.

These constants also have a direct error interpretation. Let $\boldsymbol{\theta}_\star \in \mathcal{X}$ generate (13), and let $\widehat{\boldsymbol{\theta}}$ be a global minimizer of the intensity least-squares criterion. If $\alpha_{\mathcal{X}} > 0$, then

$$\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_\star\|_2 \leq \frac{2\|\mathbf{w}\|_2}{\alpha_{\mathcal{X}}}. \tag{26}$$

Indeed, optimality gives a residual no larger than $\|\mathbf{w}\|_2$; the triangle inequality followed by the lower Lipschitz bound gives (26).

For full-channel training,

$$\mathbf{G}_{\mathrm{p}} = \mathbf{X}_{\mathrm{p}} \otimes \mathbf{I}_{N_{\mathrm{r}}}, \qquad \mathbf{X}_{\mathrm{p}} \in \mathbb{C}^{T_{\mathrm{p}}\times L}, \tag{27}$$

and therefore

$$\begin{aligned}\mathrm{rank}(\mathbf{G}_{\mathrm{p}}) &= N_{\mathrm{r}}\,\mathrm{rank}(\mathbf{X}_{\mathrm{p}}),\\ \sigma_{\min}(\mathbf{G}_{\mathrm{p}}) &= \sigma_{\min}(\mathbf{X}_{\mathrm{p}}),\\ \kappa(\mathbf{G}_{\mathrm{p}}) &= \kappa(\mathbf{X}_{\mathrm{p}}).\end{aligned} \tag{28}$$

Thus, $\mathrm{rank}(\mathbf{X}_{\mathrm{p}}) = L$ and $T_{\mathrm{p}} \geq L$ are necessary to preserve the complex channel field. They are not sufficient for the intensity mapping. With one real intensity per RIS element and pilot time, $Q = N_{\mathrm{r}}T_{\mathrm{p}}$ and $D = N_{\mathrm{r}}L$, so (24) requires

$T_{\rm p} \geq 2L$. Even when this count is met, the pilot and reference phases must make $\mathbf{J}_{\rm p}$ full column rank and well conditioned.

If the far-field model $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is used and the angles are known, the unknown can be reduced to $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_L]^T$, with

$$\mathbf{G}_\alpha = \begin{bmatrix} \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\rm p}[0]) \\ \vdots \\ \mathbf{V}\,\mathrm{diag}(\mathbf{x}_{\rm p}[T_{\rm p}-1]) \end{bmatrix}. \tag{29}$$

The necessary count becomes $N_{\rm r}T_{\rm p} \geq 2L$. In addition, $\mathbf{G}_\alpha$ must preserve every gain direction and the reference must provide two independent real projections of those directions. Closely spaced angles reduce $\sigma_{\min}(\mathbf{V})$ and hence both the field-domain and energy-domain stability margins.

D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let $\mathcal{X}_{\rm d} = \mathcal{S}^{LK_{\rm d}}$. Since this set is finite, global uniqueness and a positive minimum separation are equivalent. Define

$$d_{\min,\rm d}^{(E)} \triangleq \min_{\substack{\mathbf{s}_1,\mathbf{s}_2 \in \mathcal{X}_{\rm d} \\ \mathbf{s}_1 \neq \mathbf{s}_2}} \left\| |\mathbf{G}_{\rm d}\mathbf{s}_1 + \mathbf{b}_{\rm d}|^2 - |\mathbf{G}_{\rm d}\mathbf{s}_2 + \mathbf{b}_{\rm d}|^2 \right\|_2. \tag{30}$$

Corollary 1 (Discrete uniqueness and noise-limited reliability)**.** *The finite-alphabet symbols are uniquely identifiable in the noiseless case if and only if $d_{\min,\rm d}^{(E)} > 0$. Equivalently, for every distinct pair,*

$$\mathbf{J}_{\rm d}\left(\frac{\mathbf{s}_1 + \mathbf{s}_2}{2}\right)(\mathbf{s}_1 - \mathbf{s}_2)_{\mathbb{R}} \neq \mathbf{0}. \tag{31}$$

For an arbitrary additive perturbation, the nearest-intensity detector returns the transmitted symbol vector whenever

$$\|\mathbf{w}\|_2 < \frac{d_{\min,\rm d}^{(E)}}{2}. \tag{32}$$

Under exact CSI and additive white Gaussian post-detection noise $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_w^2\mathbf{I})$, the error probability of the binary ML comparison between distinct candidates $\mathbf{s}_i$ and $\mathbf{s}_j$ is

$$P_{i\to j} = Q\left(\frac{d_E(\mathbf{s}_i, \mathbf{s}_j)}{2\sigma_w}\right). \tag{33}$$

Consequently, because the minimum is attained on the finite candidate set,

$$P_{\rm pair,worst} \triangleq \max_{i\neq j} P_{i\to j} = Q\left(\frac{d_{\min,\rm d}^{(E)}}{2\sigma_w}\right). \tag{34}$$

If $M_{\rm c} = |\mathcal{X}_{\rm d}| = |\mathcal{S}|^{LK_{\rm d}}$ is the number of candidate symbol vectors, the union bound gives

$$P_{\rm B} \leq (M_{\rm c} - 1)Q\left(\frac{d_{\min,\rm d}^{(E)}}{2\sigma_w}\right). \tag{35}$$

Using $Q(x) \leq \frac{1}{2}e^{-x^2/2}$ further yields

$$P_{\rm B} \leq \frac{M_{\rm c}-1}{2}\exp\left[-\frac{\left(d_{\min,\rm d}^{(E)}\right)^2}{8\sigma_w^2}\right]. \tag{36}$$

Proof. Let $\boldsymbol{\mu}_k = |\mathbf{G}_{\rm d}\mathbf{s}_k + \mathbf{b}_{\rm d}|^2$. Positive separation of the finite set $\{\boldsymbol{\mu}_k\}$ proves the noiseless statement, while the triangle inequality gives the decoding radius in (32). If $\mathbf{s}_i$ is transmitted, the binary ML comparison favors $\mathbf{s}_j$ only if $\mathbf{w}^T(\boldsymbol{\mu}_j - \boldsymbol{\mu}_i) \geq d_E^2(\mathbf{s}_i, \mathbf{s}_j)/2$. The left-hand side is Gaussian with variance $\sigma_w^2 d_E^2(\mathbf{s}_i, \mathbf{s}_j)$, which gives (33). Taking the closest pair, applying the union bound over the remaining $M_{\rm c} - 1$ candidates, and using the standard Gaussian-tail inequality yield the final three bounds. □

The data-phase field matrix has the factorization

$$\mathbf{G}_{\rm d} = (\mathbf{I}_{MK_{\rm d}} \otimes \widehat{\mathbf{H}})\boldsymbol{\Pi}\mathbf{P}_{{\rm tx},d}. \tag{37}$$

When the factors have full column rank,

$$\sigma_{\min}(\mathbf{G}_{\rm d}) \geq \sigma_{\min}(\widehat{\mathbf{H}})\sigma_{\min}(\mathbf{P}_{{\rm tx},d}), \tag{38}$$

$$\sigma_{\min}(\widehat{\mathbf{H}}) \approx \sigma_{\min}(\mathbf{V}\boldsymbol{\Gamma}) \geq \sigma_{\min}(\mathbf{V})\min_l |\alpha_l|. \tag{39}$$

These inequalities quantify only preservation in the complex field domain. Theorem 1 adds the missing condition: the reference-induced real projection must also be nonzero for every constellation secant.

E. A Constructive Four-Phase Reference Bound

The preceding conditions are exact but depend on the unknown midpoint. A phase-complete reference pattern gives a simple midpoint-independent sufficient condition. Suppose a base field sample $u_q = [\mathbf{G}\boldsymbol{\theta}]_q$ is repeated with four known references

$$b_{q,k} = \rho_q e^{j\phi_k}, \qquad \phi_k \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \tag{40}$$

Denote the resulting stacked mapping by $\mathcal{T}_{\rm 4ph}(\boldsymbol{\theta})$.

Theorem 2 (Four-phase global lower bound)**.** *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$,*

$$\begin{aligned} &\|\mathcal{T}_{\rm 4ph}(\boldsymbol{\theta}_1) - \mathcal{T}_{\rm 4ph}(\boldsymbol{\theta}_2)\|_2^2 \\ &\quad = 4\left\||\mathbf{G}\boldsymbol{\theta}_1|^2 - |\mathbf{G}\boldsymbol{\theta}_2|^2\right\|_2^2 + 8\sum_q \rho_q^2 |[\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)]_q|^2. \end{aligned} \tag{41}$$

Consequently, with $\rho_{\min} = \min_q \rho_q$,

$$\|\mathcal{T}_{\rm 4ph}(\boldsymbol{\theta}_1) - \mathcal{T}_{\rm 4ph}(\boldsymbol{\theta}_2)\|_2 \geq 2\sqrt{2}\rho_{\min}\|\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\|_2. \tag{42}$$

If $\mathbf{G}$ has full column rank, then

$$\alpha_{\mathcal{X}} \geq 2\sqrt{2}\rho_{\min}\sigma_{\min}(\mathbf{G}) > 0 \tag{43}$$

for every feasible set $\mathcal{X}$. For a finite alphabet, full column rank can be replaced by $\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing$.

Proof. For one field coordinate, put $c_q = |u_{1,q}|^2 - |u_{2,q}|^2$ and $\Delta u_q = u_{1,q} - u_{2,q}$. Summing $[c_q + 2\rho_q \,\mathrm{Re}\{e^{-j\phi_k}\Delta u_q\}]^2$ over the four phases cancels all cross terms and gives $4c_q^2 + 8\rho_q^2|\Delta u_q|^2$. Summing over q proves (41); the remaining claims follow from singular value or restricted-difference bounds. □

The construction also has a transparent holographic interpretation. Opposite reference phases cancel the unknown self-intensity:

$$\mathrm{Re}\{u_q\} = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q}, \quad \mathrm{Im}\{u_q\} = \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \tag{44}$$

Thus, four phase-shifted energy measurements recover two orthogonal real projections of each complex field sample. This proves explicitly that reference phase diversity can remove the global phase ambiguity and yields the stability lower bound in (43). Fewer or hardware-constrained phases may still be unique, but their quality should be evaluated through $\alpha_{\mathcal{X}}$, $\underline{\sigma}_J$, or $d^{(E)}_{\min,\mathrm{d}}$, rather than through the rank of $\mathbf{G}$ alone.

The resulting design criteria are therefore

$$\max_{\mathbf{b}_\mathrm{p}\in\mathcal{B}_\mathrm{p}} \inf_{\boldsymbol{\theta}\in\mathcal{X}_\mathrm{p}} \sigma_{\min}[\mathbf{J}_\mathrm{p}(\boldsymbol{\theta})] \quad \text{and} \quad \max_{\mathbf{b}_\mathrm{d}\in\mathcal{B}_\mathrm{d}} d^{(E)}_{\min,\mathrm{d}} \tag{45}$$

for continuous training and discrete data detection, respectively. The first maximizes a uniform local/global lower bound, whereas the second directly maximizes the finite-alphabet decision margin.

IV. Estimation and Detection Algorithms

Based on the uniqueness and stability analysis in Section III, this section develops practical recovery algorithms for the proposed holographic RIS receiver. The key point is that the algorithms should not be chosen only according to the algebraic rank of the equivalent matrix. Instead, the training stage should operate in a region where the intensity-domain Jacobian is well conditioned, while the data detector should preserve a positive energy-domain distance between finite-alphabet candidates.

The resulting receiver follows a two-stage structure. In the training stage, the unknown variables are the DOAs and the complex path gains. They are continuous parameters, and the far-field model provides a differentiable low-dimensional representation. A damped Gauss–Newton refinement is therefore used after a coarse initialization. In the data stage, the channel estimate is fixed and the unknown vector belongs to a finite constellation. Maximum-likelihood (ML) detection gives the finite-alphabet benchmark associated with the energy-domain minimum distance, whereas a reference-assisted Gerchberg–Saxton (GS) detector provides a lower-complexity iterative implementation for repeated data blocks.

A. Overall Two-Stage Recovery Framework

The proposed receiver retains the two-phase structure of Section II. During training, the far-field model in (5) is parameterized by the real vector

$$\boldsymbol{\eta} = \begin{bmatrix}\boldsymbol{\theta}^T & \boldsymbol{\phi}^T & \mathrm{Re}\{\boldsymbol{\alpha}\}^T & \mathrm{Im}\{\boldsymbol{\alpha}\}^T\end{bmatrix}^T \in \mathbb{R}^{D_\eta}, \tag{46}$$

where $\boldsymbol{\theta}$, $\boldsymbol{\phi}$, and $\boldsymbol{\alpha}$ collect the polar angles, azimuth angles, and complex path gains, respectively; thus, $D_\eta = 4L$. Let $\mathbf{g}_\mathrm{p}(\boldsymbol{\eta}) = \mathrm{col}_t\{\mathbf{H}(\boldsymbol{\eta})\mathbf{x}_\mathrm{p}[t]\} \in \mathbb{C}^{Q_\mathrm{p}}$, where $Q_\mathrm{p} = N_\mathrm{r}T_\mathrm{p}$. The parametric training model is

$$\mathbf{z}_\mathrm{p} = |\mathbf{g}_\mathrm{p}(\boldsymbol{\eta}) + \mathbf{b}_\mathrm{p}|^2 + \mathbf{w}_\mathrm{p}, \tag{47}$$

After Gauss–Newton refinement, the receiver reconstructs $\widehat{\mathbf{H}} = \mathbf{H}(\widehat{\boldsymbol{\eta}})$ and forms $\widehat{\mathbf{G}}_\mathrm{d} = \mathbf{G}_\mathrm{d}(\widehat{\mathbf{H}})$ in (12). The training reference is designed to improve the conditioning of $\mathbf{J}_\mathrm{p}$, whereas the data reference is designed to enlarge the finite-alphabet energy distance. The following two subsections give the corresponding estimators.

B. Joint DOA and Channel-Gain Estimation

1) Nonlinear Least-Squares Formulation: Based on (47), joint DOA and channel-gain estimation is formulated as

$$\hat{\boldsymbol{\eta}} = \arg\min_{\boldsymbol{\eta}\in\Omega} F_\mathrm{p}(\boldsymbol{\eta}), \qquad F_\mathrm{p}(\boldsymbol{\eta}) \triangleq \frac{1}{2}\left\|\mathbf{z}_\mathrm{p} - |\mathbf{g}_\mathrm{p}(\boldsymbol{\eta}) + \mathbf{b}_\mathrm{p}|^2\right\|_2^2, \tag{48}$$

where Ω is the feasible parameter region. This physical parameterization uses $4L$ real unknowns rather than the $2N_\mathrm{r}L$ real entries of an unstructured channel. Its Jacobian also exposes weak gains and poorly separated DOAs through small singular values.

2) Jacobian of the Parametric Intensity Mapping: At the t-th pilot instant, the noise-free received field is given by

$$\mathbf{g}_\mathrm{p}[t] = \sum_{l=1}^{L} \alpha_l x_{\mathrm{p},l}[t]\mathbf{v}_l, \tag{49}$$

where $x_{\mathrm{p},l}[t]$ denotes the pilot symbol transmitted by the l-th user, corresponding to the l-th entry of the pilot vector $\mathbf{x}_\mathrm{p}[t]$.

Let $\boldsymbol{\psi}_l \in \mathbb{R}^{N_\mathrm{r}}$ denote the phase vector whose $\iota(n,m)$-th entry is $[\boldsymbol{\psi}_l]_{\iota(n,m)} = \psi^{(l)}_{n,m}$. Since $\mathbf{v}_l(\theta_l,\phi_l) = \exp(j\boldsymbol{\psi}_l)$, where the exponential is applied element-wise, its angular derivatives can be written as

$$\frac{\partial \mathbf{v}_l}{\partial \theta_l} = j\frac{\partial \boldsymbol{\psi}_l}{\partial \theta_l} \odot \mathbf{v}_l, \quad \frac{\partial \mathbf{v}_l}{\partial \phi_l} = j\frac{\partial \boldsymbol{\psi}_l}{\partial \phi_l} \odot \mathbf{v}_l, \tag{50}$$

where the $\iota(n,m)$-th entries are $[\partial\boldsymbol{\psi}_l/\partial\theta_l]_{\iota(n,m)} = k_0\cos\theta_l\,(y_n\cos\phi_l + z_m\sin\phi_l)$ and $[\partial\boldsymbol{\psi}_l/\partial\phi_l]_{\iota(n,m)} = k_0\sin\theta_l\,(-y_n\sin\phi_l + z_m\cos\phi_l)$.

Using (49), and defining $\alpha_{l,R} = \mathrm{Re}\{\alpha_l\}$ and $\alpha_{l,I} = \mathrm{Im}\{\alpha_l\}$, the field derivatives at pilot instant t are

$$\begin{aligned} &\frac{\partial \mathbf{g}_\mathrm{p}[t]}{\partial \theta_l} = \alpha_l x_{\mathrm{p},l}[t]\frac{\partial \mathbf{v}_l}{\partial \theta_l}, && \frac{\partial \mathbf{g}_\mathrm{p}[t]}{\partial \phi_l} = \alpha_l x_{\mathrm{p},l}[t]\frac{\partial \mathbf{v}_l}{\partial \phi_l}, \\ &\frac{\partial \mathbf{g}_\mathrm{p}[t]}{\partial \alpha_{l,R}} = x_{\mathrm{p},l}[t]\mathbf{v}_l, && \frac{\partial \mathbf{g}_\mathrm{p}[t]}{\partial \alpha_{l,I}} = jx_{\mathrm{p},l}[t]\mathbf{v}_l. \end{aligned} \tag{51}$$

The derivatives over all pilot instants are stacked in the same order as $\mathbf{g}_\mathrm{p}(\boldsymbol{\eta})$. Thus, the Jacobian of the parametric intensity mapping is

$$\mathbf{J}_\mathrm{p}(\boldsymbol{\eta}) = 2\,\mathrm{Re}\left\{\mathrm{diag}\big(\mathbf{u}^*_\mathrm{p}(\boldsymbol{\eta})\big)\,\frac{\partial \mathbf{g}_\mathrm{p}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^\mathsf{T}}\right\} \in \mathbb{R}^{Q_\mathrm{p}\times D_\eta}. \tag{52}$$

The same Jacobian determines both the local identifiability discussed in Section III and the numerical behavior of the Gauss–Newton iterations. In particular, a small $\sigma_{\min}(\mathbf{J}_\mathrm{p})$ indicates that certain perturbations in the DOAs or path gains produce only weak intensity variations, leading to both poor estimation stability and poorly conditioned Gauss–Newton updates. This is the reason for using phase-diverse references during training: the reference term changes the real projection directions in the Jacobian and can convert a nearly invisible complex-field perturbation into a measurable intensity perturbation.

3) Damped Gauss–Newton Update: At iteration i, define $\mathbf{u}_{\mathrm{p}}^{(i)} = \mathbf{g}_{\mathrm{p}}\left(\boldsymbol{\eta}^{(i)}\right) + \mathbf{b}_{\mathrm{p}}$, $\mathbf{e}_{\mathrm{p}}^{(i)} = \mathbf{z}_{\mathrm{p}} - \left|\mathbf{u}_{\mathrm{p}}^{(i)}\right|^2$, and $\mathbf{J}_{\mathrm{p}}^{(i)} = \mathbf{J}_{\mathrm{p}}(\boldsymbol{\eta}^{(i)})$ denotes the intensity-domain Jacobian evaluated at the current iterate.

The damped Gauss–Newton increment is then computed as

$$\Delta\boldsymbol{\eta}^{(i)} = \left[\left(\mathbf{J}_{\mathrm{p}}^{(i)}\right)^T \mathbf{J}_{\mathrm{p}}^{(i)} + \lambda^{(i)}\mathbf{I}\right]^{-1} \left(\mathbf{J}_{\mathrm{p}}^{(i)}\right)^T \mathbf{e}_{\mathrm{p}}^{(i)}, \tag{53}$$

where $\lambda^{(i)} \geq 0$ is a damping parameter introduced to improve numerical stability when the Jacobian is poorly conditioned. The parameter vector is updated as

$$\boldsymbol{\eta}_{\mathrm{trial}}^{(i+1)} = \Pi_{\Omega}\left(\boldsymbol{\eta}^{(i)} + \Delta\boldsymbol{\eta}^{(i)}\right), \tag{54}$$

where $\Pi_{\Omega}(\cdot)$ projects the angular parameters onto the feasible region. The trial point is accepted and the damping is reduced when it lowers F_{p}; otherwise, the current iterate is retained and the damping is increased.

4) Initialization: Because (48) is nonconvex, the complex field is first estimated coarsely and then fitted to the far-field model. With the four-phase training reference, opposite-phase differences in (44) directly give the field samples. Stacking them gives $\hat{\mathbf{g}}_{\mathrm{p}}$. A coarse unstructured channel estimate is then obtained by least squares as

$$\mathbf{h}^{(0)} = \mathbf{G}_{\mathrm{p}}^{\dagger}\hat{\mathbf{g}}_{\mathrm{p}}, \qquad \mathbf{H}^{(0)} = \mathrm{unvec}\left(\mathbf{h}^{(0)}\right). \tag{55}$$

When four separate reference phases are unavailable, the reference-dominated approximation can be used to obtain

$$\mathbf{h}^{(0)} = \arg\min_{\mathbf{h}} \left\|\mathbf{z}_{\mathrm{p}} - |\mathbf{b}_{\mathrm{p}}|^2 - 2\,\mathrm{Re}\left\{\mathbf{b}_{\mathrm{p}}^* \odot \mathbf{G}_{\mathrm{p}}\mathbf{h}\right\}\right\|_2^2, \tag{56}$$

from which $\mathbf{H}^{(0)}$ is reconstructed.

The initial DOA of each user is then obtained through a coarse angular search,

$$\left(\theta_l^{(0)}, \phi_l^{(0)}\right) = \arg\max_{(\theta,\phi)\in\mathcal{G}_{\Omega}} \frac{\left|\mathbf{v}^H(\theta,\phi)\mathbf{h}_l^{(0)}\right|^2}{\|\mathbf{v}(\theta,\phi)\|_2^2}, \tag{57}$$

where $\mathcal{G}_{\Omega}$ denotes a coarse angular grid. Given the coarse DOA estimate, the corresponding complex path gain is obtained by least-squares projection onto the steering vector. These coarse parameter estimates are then used to initialize the damped Gauss–Newton iterations.

The complete joint estimation procedure is summarized in Algorithm 1.

The Gauss–Newton method is a local optimization procedure and does not guarantee convergence to the global minimizer from an arbitrary initialization. Its use is justified here by the differentiable low-dimensional channel parameterization, the availability of pilot observations, and the possibility of obtaining a coarse angular initialization before iterative refinement.

C. Finite-Alphabet Symbol Detection

After training, $\widehat{\mathbf{H}}$ is used to construct $\widehat{\mathbf{G}}_{\mathrm{d}}$ in (12). Because $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{D_{\mathrm{s}}}$ is finite-alphabet, reliability is governed by the energy-domain separation in (30), rather than by a continuous Jacobian rank condition.

Algorithm 1 Joint DOA and Channel-Gain Estimation

Require: Training observations $\mathbf{z}_{\mathrm{p}}$, pilots $\mathbf{x}_{\mathrm{p}}[t]$, reference vector $\mathbf{b}_{\mathrm{p}}$, RIS geometry, feasible parameter region Ω, maximum iteration number I_{c}, initial damping factor $\lambda^{(0)}$, and constants $\epsilon_{\mathrm{c}}, \epsilon_0 > 0$.
Ensure: Estimated parameter vector $\hat{\boldsymbol{\eta}}$ and channel $\hat{\mathbf{H}}$.

```
Obtain H^(0) from the four-phase field estimate when available; otherwise use (56).
Obtain coarse DOAs using (57).
Form η^(0) according to (46).
for i = 0, 1, ..., I_c − 1 do
    Compute u_p^(i) = g_p(η^(i)) + b_p.
    Compute the residual e_p^(i) = z_p − |u_p^(i)|^2.
    Construct J_p^(i) using (52).
    Compute Δη^(i) using (53).
    Compute the projected trial point using (54).
    if the trial point decreases F_p then
        Accept η^(i+1) = η_trial^(i+1).
        Reduce the damping parameter.
    else
        Set η^(i+1) = η^(i).
        Increase the damping parameter.
    end if
    if ||Δη^(i)||_2 / (||η^(i)||_2 + ε_0) < ε_c then
        break
    end if
end for
Set η̂ = η^(i+1).
Reconstruct Ĥ = H(η̂).
```

1) Maximum-Likelihood Detection: Assuming independent Gaussian post-detection noise with identical variance, the maximum-likelihood detector is equivalent to

$$\hat{\mathbf{s}}_{\mathrm{ML}} = \arg\min_{\mathbf{s}\in\mathcal{S}^{D_{\mathrm{s}}}} \left\|\mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2\right\|_2^2. \tag{58}$$

The corresponding estimated minimum distance is obtained from (30) by replacing $\mathbf{G}_{\mathrm{d}}$ with $\widehat{\mathbf{G}}_{\mathrm{d}}$. A positive value ensures distinct noiseless intensity vectors, but exhaustive ML evaluates $|\mathcal{S}|^{D_{\mathrm{s}}}$ candidates and is therefore used mainly as a short-block benchmark.

2) Linearized Initialization and Candidate Screening: If the reference-induced interference dominates the unknown self-intensity, $\mathbf{z}_{\mathrm{d}} - |\mathbf{b}_{\mathrm{d}}|^2 \approx \mathbf{A}_{\mathrm{d}}\bar{\mathbf{s}}_{\mathrm{d}}$, where $\bar{\mathbf{s}} = [\mathrm{Re}\{\mathbf{s}\}^T, \mathrm{Im}\{\mathbf{s}\}^T]^T$ and

$$\mathbf{A}_{\mathrm{d}} = 2\left[\mathrm{Re}\{\mathrm{diag}(\mathbf{b}_{\mathrm{d}}^*)\hat{\mathbf{G}}_{\mathrm{d}}\} \quad -\mathrm{Im}\{\mathrm{diag}(\mathbf{b}_{\mathrm{d}}^*)\hat{\mathbf{G}}_{\mathrm{d}}\}\right]. \tag{59}$$

A regularized real-domain initialization is

$$\bar{\mathbf{s}}_{\mathrm{lin}} = \left(\mathbf{A}_{\mathrm{d}}^T\mathbf{A}_{\mathrm{d}} + \gamma_{\mathrm{lin}}\mathbf{I}\right)^{-1}\mathbf{A}_{\mathrm{d}}^T\left(\mathbf{z}_{\mathrm{d}} - |\mathbf{b}_{\mathrm{d}}|^2\right), \tag{60}$$

where $\gamma_{\mathrm{lin}} \geq 0$. The complex estimate $[\mathbf{I}; j\mathbf{I}]\bar{\mathbf{s}}_{\mathrm{lin}}$ is projected element-wise onto $\mathcal{S}$ and used either to initialize an iterative detector or to center a reduced ML candidate list.

Algorithm 2 Reference-Assisted Gerchberg–Saxton Symbol Detection

Require: Data observations $\mathbf{z}_\mathrm{d}$, matrix $\hat{\mathbf{G}}_\mathrm{d}$, reference vector $\mathbf{b}_\mathrm{d}$, constellation $\mathcal{S}$, regularization factor γ, maximum iteration number I_d, and constants $\epsilon_\mathrm{d}, \epsilon_0 > 0$.
Ensure: Detected symbol vector $\hat{\mathbf{s}}_\mathrm{d}$.
Compute $\mathbf{r}_\mathrm{d} = \sqrt{[\mathbf{z}_\mathrm{d}]_+}$.
Precompute $\mathbf{B}_\mathrm{d}$ according to (63).
Initialize $\mathbf{s}^{(0)}$ using a linearized estimate, the previous data block, or a random constellation vector.
for $i = 0, 1, \ldots, I_\mathrm{d} - 1$ **do**
  Compute the predicted affine field using (61).
  Apply the magnitude constraint using (62).
  Back-project using (64).
  Apply the constellation projection using (65).
  if $\|\mathbf{s}^{(i+1)} - \mathbf{s}^{(i)}\|_2 / \big(\|\mathbf{s}^{(i)}\|_2 + \epsilon_0\big) < \epsilon_\mathrm{d}$ **then**
    break
  end if
end for
Output $\hat{\mathbf{s}}_\mathrm{d} = \mathcal{Q}_\mathcal{S}\big(\mathbf{s}^{(i+1)}\big)$.

3) Reference-Assisted Gerchberg–Saxton Detection: The conventional Gerchberg–Saxton algorithm alternates between a measurement-magnitude constraint and a signal-domain constraint. The detector considered here is not a direct application of the conventional homogeneous phase-retrieval algorithm. Instead, it is adapted to the affine observation model by including the known reference vector in the forward projection and subtracting it before the signal-domain back-projection.

Given the current symbol estimate $\mathbf{s}^{(i)}$, the predicted affine field is

$$\mathbf{u}_\mathrm{d}^{(i)} = \hat{\mathbf{G}}_\mathrm{d}\mathbf{s}^{(i)} + \mathbf{b}_\mathrm{d}. \tag{61}$$

The measured-magnitude constraint is then enforced as

$$\tilde{\mathbf{u}}_\mathrm{d}^{(i)} = \sqrt{[\mathbf{z}_\mathrm{d}]_+} \odot \exp\Big(j\angle \mathbf{u}_\mathrm{d}^{(i)}\Big), \tag{62}$$

where $[\mathbf{z}]_+ = \max(\mathbf{z}, \mathbf{0})$ is applied element-wise.

The reference vector is then removed, and the resulting field is projected back onto the signal domain. To improve numerical stability when $\hat{\mathbf{G}}_\mathrm{d}$ is ill-conditioned, define the regularized back-projection matrix

$$\mathbf{B}_\mathrm{d} = \big(\hat{\mathbf{G}}_\mathrm{d}^\mathsf{H}\hat{\mathbf{G}}_\mathrm{d} + \gamma\mathbf{I}\big)^{-1}\hat{\mathbf{G}}_\mathrm{d}^\mathsf{H}, \tag{63}$$

where $\gamma \geq 0$ is a regularization parameter. When $\gamma = 0$ and $\hat{\mathbf{G}}_\mathrm{d}$ has full column rank, $\mathbf{B}_\mathrm{d} = \hat{\mathbf{G}}_\mathrm{d}^\dagger$.

The continuous signal-domain update is

$$\tilde{\mathbf{s}}^{(i+1)} = \mathbf{B}_\mathrm{d}\big(\tilde{\mathbf{u}}_\mathrm{d}^{(i)} - \mathbf{b}_\mathrm{d}\big). \tag{64}$$

The finite-alphabet structure is incorporated through

$$\mathbf{s}^{(i+1)} = \mathcal{Q}_\mathcal{S}\big(\tilde{\mathbf{s}}^{(i+1)}\big), \tag{65}$$

where $\mathcal{Q}_\mathcal{S}(\cdot)$ denotes element-wise nearest-neighbor quantization onto the constellation $\mathcal{S}$.

In the proposed detector, (65) is applied after every back-projection to exploit the finite-alphabet prior. The complete procedure is summarized in Algorithm 2.

The proposed reference-assisted GS detector is attractive for repeated data detection because $\mathbf{B}_\mathrm{d}$ can be precomputed whenever the estimated channel remains unchanged over a coherence block. Each subsequent iteration then mainly involves matrix-vector multiplications and element-wise operations.

D. Computational Complexity and Implementation Considerations

Let I_c and I_d denote the numbers of channel-estimation and symbol-detection iterations, respectively. The principal computational costs are summarized in Table I.

For joint DOA and gain estimation, the dominant operations are forming $\mathbf{J}_\mathrm{p}^T\mathbf{J}_\mathrm{p}$ and solving a $D_\eta \times D_\eta$ system. Since $D_\eta = 4L$, this dimension is independent of N_r and is much smaller than the $2N_\mathrm{r}L$-dimensional unstructured alternative. For GS, forming $\mathbf{B}_\mathrm{d}$ costs $\mathcal{O}(Q_\mathrm{d}D_\mathrm{s}^2 + D_\mathrm{s}^3)$ once per coherence block; subsequent iterations require only matrix–vector and element-wise operations. The linearized estimate or the preceding data block can also provide a deterministic warm start.

E. Cramér–Rao Lower Bound for DOA Estimation

The CRLB follows directly from the parametric training model in (47) and its Jacobian in (52). Assume that the effective post-detection noise is white Gaussian, $\mathbf{w}_\mathrm{p} \sim \mathcal{N}(\mathbf{0}, \sigma_\mathrm{p}^2\mathbf{I})$, with variance independent of the unknown parameters. Repartition the real parameter vector in (46) as $\boldsymbol{\eta} = [\boldsymbol{\psi}^\mathsf{T}, \boldsymbol{\nu}^\mathsf{T}]^\mathsf{T}$, where $\boldsymbol{\psi} = [\boldsymbol{\theta}^\mathsf{T}, \boldsymbol{\phi}^\mathsf{T}]^\mathsf{T}$ contains the DOAs and $\boldsymbol{\nu} = [\mathrm{Re}\{\boldsymbol{\alpha}\}^\mathsf{T}, \mathrm{Im}\{\boldsymbol{\alpha}\}^\mathsf{T}]^\mathsf{T}$ contains the nuisance path gains. The Fisher information matrix (FIM) is then

$$\mathbf{F}(\boldsymbol{\eta}) = \frac{1}{\sigma_\mathrm{p}^2}\mathbf{J}_\mathrm{p}^\mathsf{T}(\boldsymbol{\eta})\mathbf{J}_\mathrm{p}(\boldsymbol{\eta}), \tag{66}$$

with the column partition $\mathbf{J}_\mathrm{p} = [\mathbf{J}_\psi \ \mathbf{J}_\nu]$. Eliminating the unknown gains gives the equivalent FIM for the DOAs,

$$\begin{aligned} \mathbf{F}_\mathrm{DOA} &= \mathbf{F}_{\psi\psi} - \mathbf{F}_{\psi\nu}\mathbf{F}_{\nu\nu}^\dagger\mathbf{F}_{\nu\psi} \\ &= \frac{1}{\sigma_\mathrm{p}^2}\mathbf{J}_\psi^\mathsf{T}\mathbf{P}_\nu^\perp\mathbf{J}_\psi, \\ \mathbf{P}_\nu^\perp &= \mathbf{I} - \mathbf{J}_\nu\mathbf{J}_\nu^\dagger, \end{aligned} \tag{67}$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose inverse. Thus, only the component of an angle-induced intensity perturbation orthogonal to the gain perturbation subspace carries DOA information. If $\mathbf{F}_\mathrm{DOA}$ is nonsingular, any unbiased DOA estimator satisfies

$$\mathrm{Cov}(\hat{\boldsymbol{\psi}}) \succeq \mathbf{F}_\mathrm{DOA}^{-1}. \tag{68}$$

The diagonal entries give the individual bounds for θ_l and ϕ_l. Thus, weak or rank-deficient directions of $\mathbf{P}_\nu^\perp\mathbf{J}_\psi$ respectively produce large or unbounded DOA limits, linking reference design directly to attainable angular accuracy.

V. Simulation Results

Numerical results evaluate channel estimation, symbol detection under perfect and estimated channel state information (CSI), and the effects of reference and RIS measurement diversity.

TABLE I
COMPUTATIONAL COMPLEXITY OF THE MAIN RECOVERY ALGORITHMS

Method	Unknown dimension	Approximate complexity	Primary role
Joint damped GN	D_η	$\mathcal{O}\big(I_\mathrm{c}\,(Q_\mathrm{p}D_\eta^2 + D_\eta^3)\big)$	DOA and channel-gain estimation
Exhaustive ML	D_s	$\mathcal{O}\big(\lvert\mathcal{S}\rvert^{D_\mathrm{s}}Q_\mathrm{d}D_\mathrm{s}\big)$	Performance benchmark
Linearized detector	D_s	$\mathcal{O}\big(Q_\mathrm{d}D_\mathrm{s}^2 + D_\mathrm{s}^3\big)$	Initialization and candidate screening
Reference-assisted GS	D_s	$\mathcal{O}(I_\mathrm{d}Q_\mathrm{d}D_\mathrm{s})$	Low-complexity symbol detection

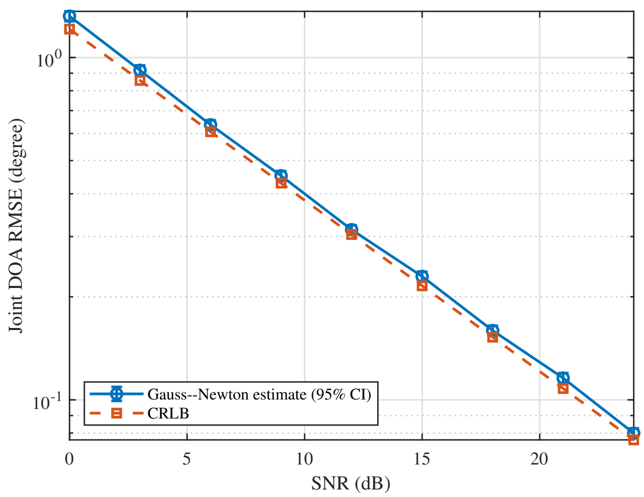

Fig. 2. Joint DOA RMSE and the corresponding CRLB versus SNR.

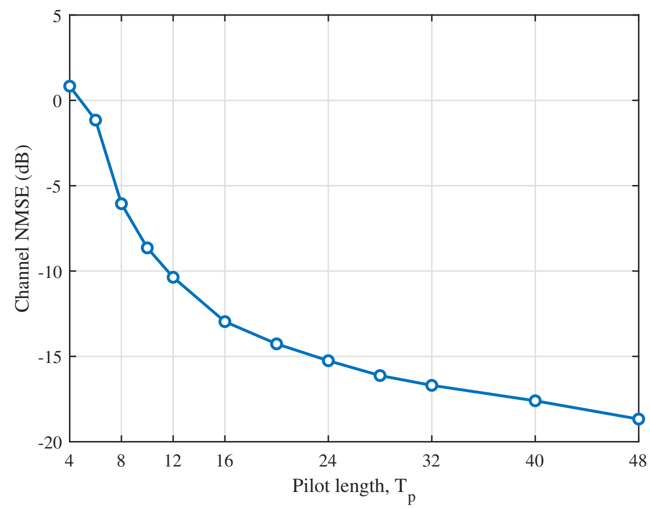

Fig. 3. Channel-estimation NMSE versus pilot length at an SNR of 12 dB.

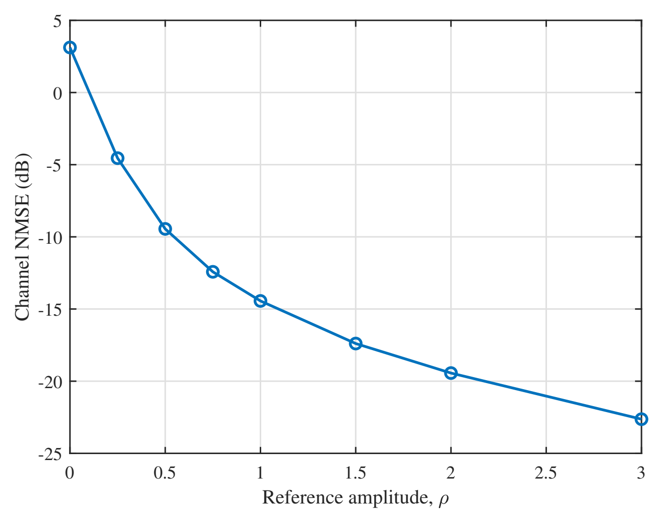

Fig. 4. Channel-estimation NMSE versus reference amplitude for fixed post-detection noise variance.

Unless otherwise stated, the simulations follow the two-phase model in Section II with a two-user 8×8 holographic RIS operating at 3.5 GHz and half-wavelength element spacing. The complex path gains are $\alpha_1 = e^{j0.35}$ and $\alpha_2 = 0.85e^{-j0.55}$, normalized QPSK is used, and the reference amplitude is 1.5. Training and data observations follow (47) and (12) with independent real Gaussian post-detection noise of variance σ_w^2 per intensity sample. For a noiseless intensity vector $\boldsymbol{\mu} = |\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b}|^2 \in \mathbb{R}^Q$, the SNR is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mathbb{E}\{\|\boldsymbol{\mu}\|_2^2\}}{Q\sigma_w^2}\right). \tag{69}$$

Scenario-specific DOAs, pilot lengths, modulation orders, and reference or RIS configurations are stated with the corresponding experiments.

A. Channel Estimation Performance

Unless stated otherwise, this subsection uses $T_\mathrm{p} = 32$ QPSK pilot observations and user DOAs $(15^\circ, 8^\circ)$ and $(35^\circ, -10^\circ)$, and the path-gain magnitudes are 1 and 0.85, respectively. Each point is averaged over 30 independent noise realizations unless noted otherwise. The reference phase pattern is selected from a 32-entry codebook by maximizing the minimum singular value of the training Jacobian.

Based on 1000 noise realizations, Fig. 2 shows that the joint DOA RMSE decreases from 1.32° at 0 dB to 0.08° at 24 dB and closely follows the CRLB. Because the constrained nonlinear estimator can be biased at finite SNR, the ordinary unbiased-estimator CRLB is used as a local accuracy benchmark rather than a strict finite-sample bound.

At 12 dB, Fig. 3 shows an NMSE improvement from 0.83 dB at the counting limit $T_\mathrm{p} = 4$ to -18.67 dB at $T_\mathrm{p} = 48$. Additional pilot diversity improves Jacobian conditioning, with diminishing returns in the overdetermined regime.

In Fig. 4, the post-detection noise variance is fixed at its nominal $\rho = 1.5$ value. The reference lowers the NMSE from 3.12 dB at $\rho = 0$ to -17.39 dB at $\rho = 1.5$ by strengthening the interference term. This idealized trend excludes saturation and quantization and therefore does not imply unbounded hardware gains.

Figure 5, averaged over 300 paired trials, compares equal-

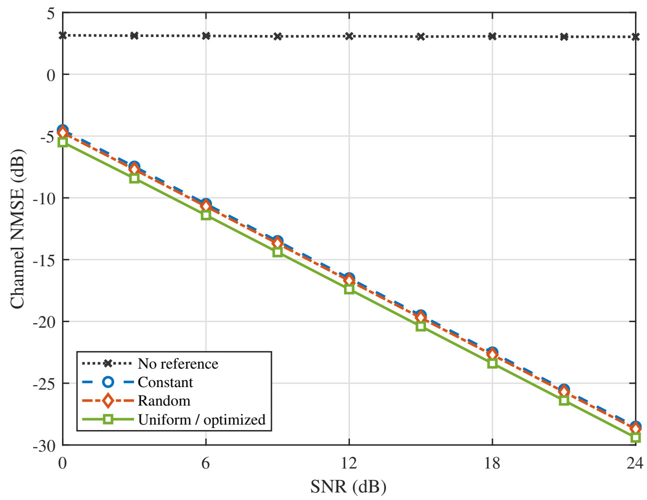

Fig. 5. Channel-estimation NMSE versus SNR for different training-reference designs.

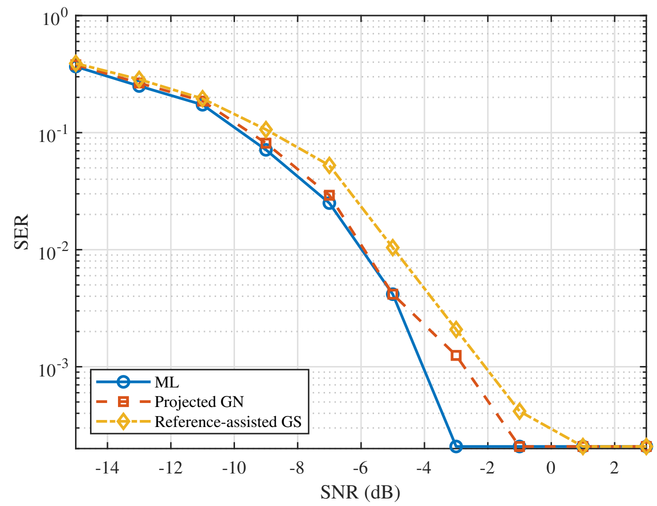

Fig. 6. SER of different symbol detectors under perfect CSI.

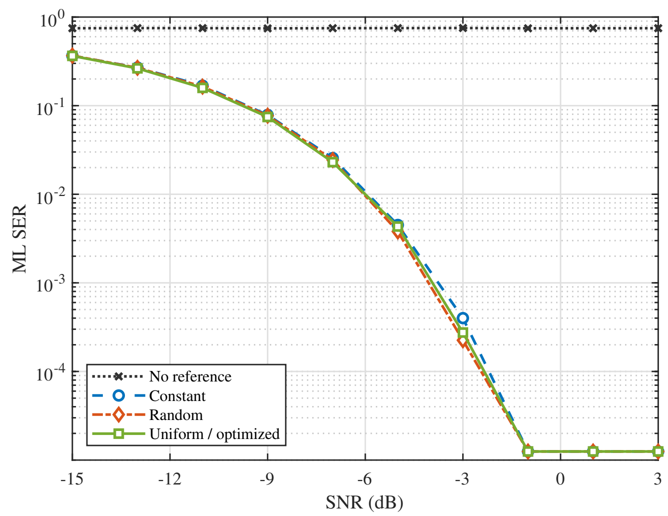

Fig. 7. ML SER versus SNR for different data-phase reference designs under perfect CSI. Zero-error points are displayed at the half-event level.

power reference patterns. Without a reference, the singular training Jacobian ($\sigma_{\min} = 3.69 \times 10^{-16}$) produces a persistent 3 dB error floor. The reference-assisted curves instead decrease with SNR; the uniform pattern gives the largest minimum singular value (9.95) and is also selected by the codebook search. Thus, the reference removes the phase ambiguity, while its phase design further improves conditioning.

B. Symbol Detection Performance with Perfect CSI

We first isolate the data detector from channel-estimation errors by constructing $\mathbf{G}_{\mathrm{d}}$ from the true channel. Unless stated otherwise, two QPSK users at the DOAs used in the preceding subsection, four reference states, and a reference amplitude of 1.5 are considered.

Figure 6 compares the detectors under the codebook-selected reference. Projected GN closely follows the ML benchmark, whereas GS incurs a moderate alternating-projection loss but retains the same waterfall trend; at -5 dB their SERs are 4.17×10^{-3}, 4.17×10^{-3}, and 1.04×10^{-2}, respectively.

Figure 7 compares equal-power references under the energy-domain SNR in (69). Without a reference, $d_{\min}^{(E)} = 0$ and the QPSK SER remains near 0.75. All reference-assisted designs remove this ambiguity. The uniform pattern has the largest normalized minimum distance (9.79) and is selected by the codebook search. Each point contains 40000 user-symbol decisions; zero-error points are plotted at the half-event level 1.25×10^{-5}.

To test whether this distance predicts more than the five representative designs, Fig. 8 uses a 32-entry equal-power reference codebook with different phase and spatial-support patterns. At -6 dB, the correlation between the normalized minimum distance and the logarithm of the ML SER is -0.98. Thus, increasing the worst-case intensity separation systematically lowers the detection error, directly supporting the max–min criterion in (45).

Finally, Fig. 9 uses the same optimized reference to compare QPSK, 16-QAM, 64-QAM, and 256-QAM. As the modulation order increases, the denser constellation reduces the separation between intensity codewords and shifts the SER curve toward higher SNR. Over the sampled SNR grid, the SER first falls below 10^{-2} at -3, 3, 9, and 15 dB for QPSK, 16-QAM, 64-QAM, and 256-QAM, respectively. Thus, the reference removes the structural phase ambiguity for all four constellations, whereas the remaining SNR requirement is governed by finite-alphabet energy separation.

C. End-to-End Symbol Detection with Estimated CSI

Figure 10 constructs $\mathbf{G}_{\mathrm{d}}$ from the estimated channel and uses equal training and data SNRs. At -6 dB, estimated-CSI SERs are about 0.2, compared with roughly 10^{-2} under perfect CSI, and the gap narrows as the channel NMSE improves. Channel mismatch, rather than the choice among the three detectors, therefore dominates at low SNR.

Figure 11 separates the two stages. With data SNR fixed at -6 dB, increasing the training SNR lowers the reference-assisted SER toward the data-noise floor, whereas the no-

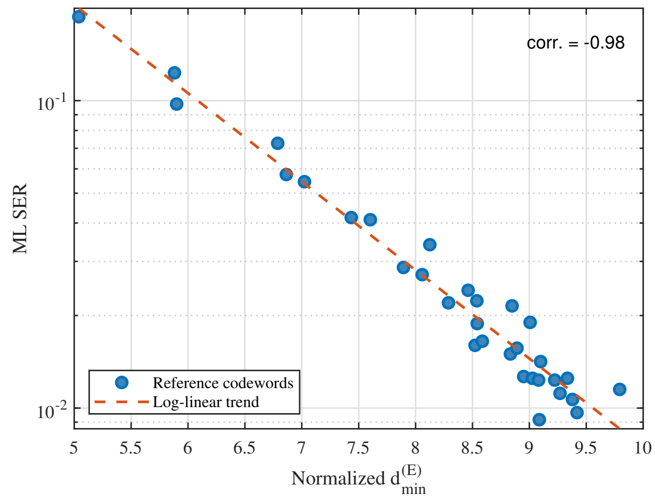

Fig. 8. ML SER versus normalized energy-domain minimum distance over an equal-power reference codebook.

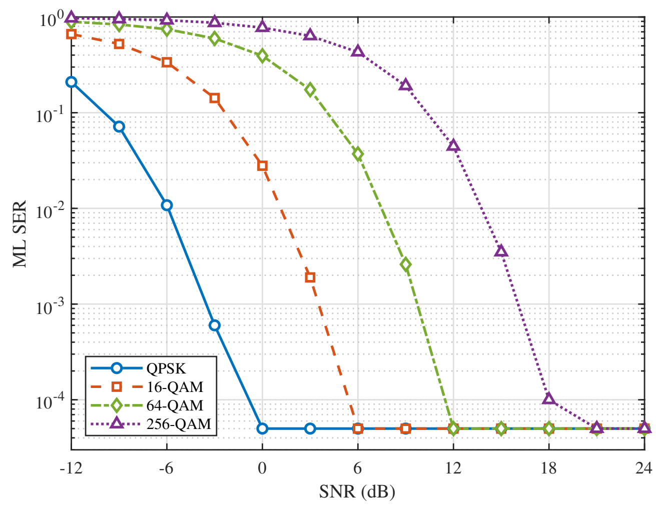

Fig. 9. ML SER for QPSK, 16-QAM, 64-QAM, and 256-QAM under perfect CSI. Zero-error points are displayed at the half-event level.

reference case remains near 0.75. With training SNR fixed at 12 dB, the same error floor persists without a data reference, while all phase-diverse designs show the expected waterfall. Thus, the training reference enables CSI recovery and the data reference ensures finite-alphabet separability.

D. Effect of Reference and RIS Measurement Diversity

We finally examine how spatial and reference-state diversity change the finite-alphabet decision margin. To compare configurations with different numbers of observations, $d_{\min}^{(E)}$ is normalized by the root-mean-square noiseless intensity. The two-user QPSK setting and four-phase reference are used unless otherwise stated.

At -12 dB, Fig. 12 shows that increasing N_{r} from 4 to 1024 raises the normalized minimum distance from 1.68 to 39.24 and drives the ML SER toward zero. This close correspondence confirms that the element outputs provide useful spatial diversity.

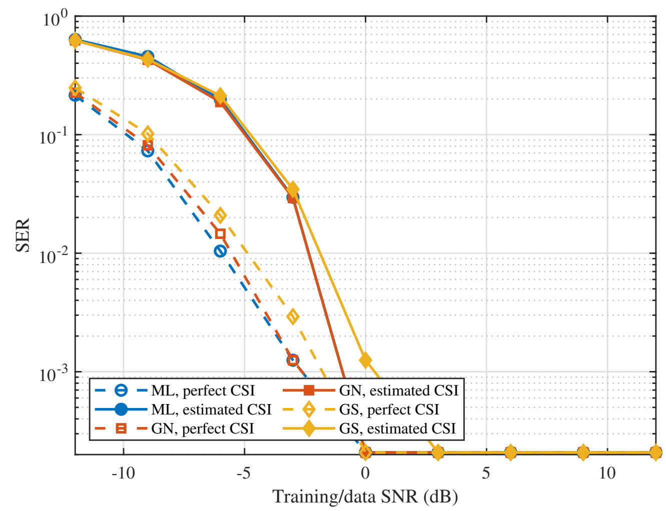

Fig. 10. SER under perfect and estimated CSI for different symbol detectors.

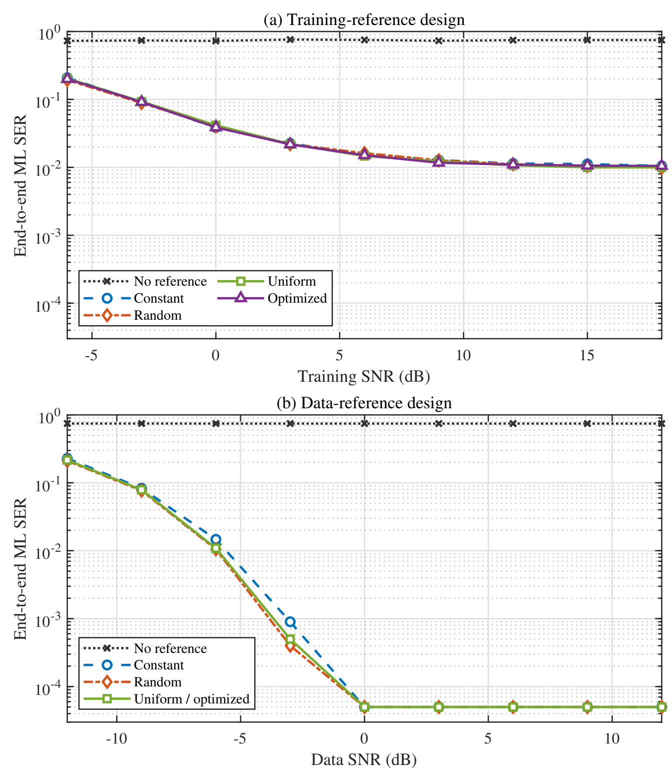

Fig. 11. End-to-end ML SER versus (a) training SNR for different training-reference designs at a data SNR of -6 dB and (b) data SNR for different data-reference designs at a training SNR of 12 dB. Zero-error points are displayed at the half-event level.

Reference-state diversity gives the analogous trend in Fig. 13. Uniformly spacing R phases over $[0, 2\pi)$ increases the normalized distance from 2.75 to 19.21 and lowers the -12 dB SER from 0.501 to 0.0111 as R grows from 1 to 16, at the cost of linearly increasing measurement and switching overhead.

Figure 14 connects spatial conditioning to end-to-end detection. Increasing the separation from $0.25°$ to $8°$ raises the normalized minimum singular value from 0.0214 to 0.628 and

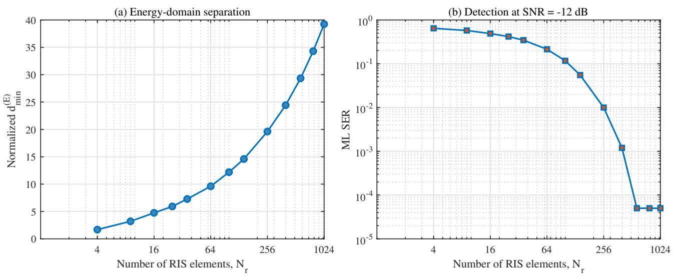

Fig. 12. Effect of the number of RIS element-level observations on (a) the normalized energy-domain minimum distance and (b) ML detection at -12 dB. Zero-error points are displayed at the half-event level.

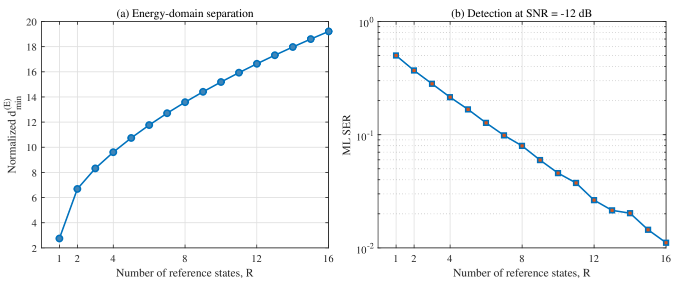

Fig. 13. Effect of the number of phase-shifted reference states on (a) the normalized energy-domain minimum distance and (b) ML detection.

lowers the estimated-CSI SER from 0.259 to 2.57×10^{-2}. The small rebound beyond 16° reflects finite-aperture sidelobes. Here the training and data SNRs are 12 and -6 dB, respectively, with $T_{\mathrm{p}} = 16$.

Finally, Fig. 15 compares equal-overhead designs with 64 samples. Selecting state-dependent element masks from a 4096-entry codebook increases the normalized minimum distance from 4.75 to 4.84. Both ML and GS exhibit the same modest but consistent improvement, showing that RIS sampling can complement reference-phase diversity.

These results support the two-part design principle in Section III: the RIS measurement matrix must first preserve relevant field differences, after which reference-phase diversity projects those differences into separated intensity observations.

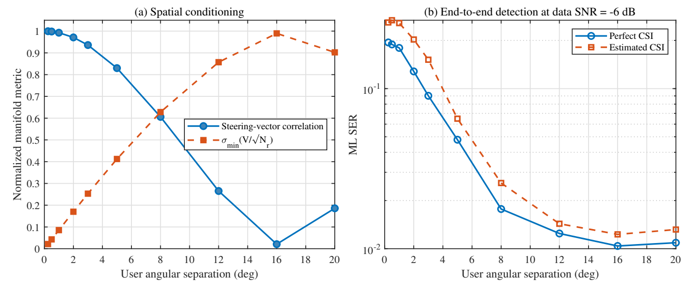

Fig. 14. Effect of user angular separation on (a) array-manifold conditioning and (b) end-to-end ML detection.

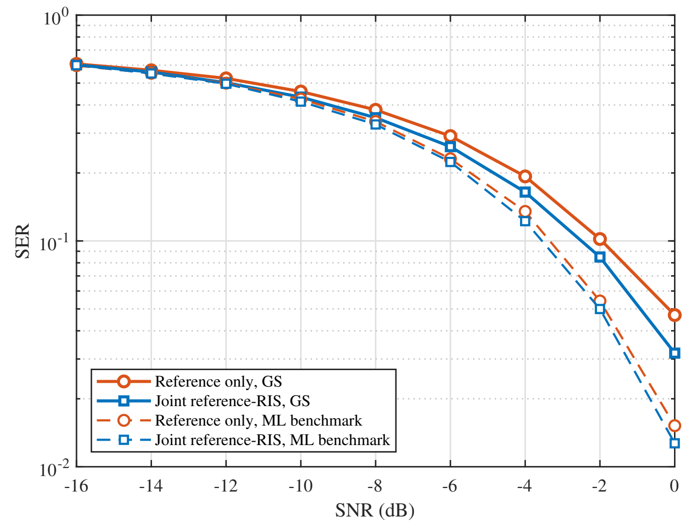

Fig. 15. SER of the proposed reference-assisted GS detector and the ML benchmark under equal-overhead reference-only and joint reference–RIS measurement diversity.

VI. Conclusion

This paper developed a reference-assisted holographic RIS receiver that recovers complex channel and data information from intensity-only measurements. The exact secant identity and energy-domain minimum distance connected continuous channel identifiability, finite-alphabet uniqueness, and noise robustness, while the four-phase construction provided an explicit stability guarantee. A damped Gauss–Newton estimator and a reference-assisted GS detector implemented the training and data stages, respectively, with the CRLB and ML detector serving as benchmarks. Simulations confirmed that reference phase design, pilot diversity, array size, and angular separation jointly determine estimation and detection performance. Future work will incorporate hardware-dependent noise, quantization and dynamic-range constraints, and frequency-selective channels.

References

[1] Q. Wu and R. Zhang, “Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming,” *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.

[2] B. Di, H. Zhang, L. Song, Y. Li, and G. Y. Li, “Reconfigurable intelligent surface-based wireless communications: Antenna design, prototyping, and experimental results,” *IEEE Access*, vol. 7, pp. 90 013–90 023, 2019.

[3] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Commun. Mag.*, vol. 58, no. 1, pp. 106–112, 2020.

[4] E. Basar, M. D. Renzo, J. de Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, "Wireless communications through reconfigurable intelligent surfaces," *IEEE Access*, vol. 7, pp. 116 753–116 773, 2019.

[5] C. Huang, A. Zappone, G. C. Alexandropoulos, M. Debbah, and C. Yuen, "Reconfigurable intelligent surfaces for energy efficiency in wireless communication," *IEEE Transactions on Wireless Communications*, vol. 18, no. 8, pp. 4157–4170, Aug. 2019.

[6] M. D. Renzo, A. Zappone, M. Debbah, M.-S. Alouini, C. Yuen, J. de Rosny, and S. Tretyakov, "Smart radio environments empowered by reconfigurable intelligent surfaces: How it works, state of research, and the road ahead," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 11, pp. 2450–2525, Nov. 2020.

[7] C. Pan, H. Ren, K. Wang, J. F. Kolb, M. Elkashlan, M. Chen, M. D. Renzo, Y. Hao, J. Wang, A. L. Swindlehurst, X. You, and L. Hanzo, "Reconfigurable intelligent surfaces for 6G systems: Principles, applications, and research directions," *IEEE Communications Magazine*, vol. 59, no. 6, pp. 14–20, Jun. 2021.

[8] E. Björnson, H. Wymeersch, B. Matthiesen, P. Popovski, L. Sanguinetti, and E. de Carvalho, "Reconfigurable intelligent surfaces: A signal processing perspective with wireless applications," *IEEE Signal Processing Magazine*, vol. 39, no. 2, pp. 135–158, Mar. 2022.

[9] M. D. Renzo, F. H. Danufane, and S. Tretyakov, "Communication models for reconfigurable intelligent surfaces: From surface electromagnetics to wireless networks optimization," *Proceedings of the IEEE*, vol. 110, no. 9, pp. 1164–1209, Sep. 2022.

[10] S. Hu, F. Rusek, and O. Edfors, "Beyond massive MIMO: The potential of data transmission with large intelligent surfaces," *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2746–2758, May 2018.

[11] D. Dardari, "Communicating with large intelligent surfaces: Fundamental limits and models," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 11, pp. 2526–2537, Nov. 2020.

[12] C. Huang, S. Hu, G. C. Alexandropoulos, A. Zappone, C. Yuen, R. Zhang, M. D. Renzo, and M. Debbah, "Holographic MIMO surfaces for 6G wireless networks: Opportunities, challenges, and trends," *IEEE Wireless Communications*, vol. 27, no. 5, pp. 118–125, Oct. 2020.

[13] H. Zhang, S. Zeng, B. Di, Y. Tan, M. D. Renzo, and L. Song, "Holographic radio surfaces: From theory to realization," *IEEE Wireless Communications*, vol. 28, no. 4, pp. 20–26, Aug. 2021.

[14] J. An, C. Yuen, Y. L. Guan, M. Di Renzo, M. Debbah, H. V. Poor, and L. Hanzo, "Stacked intelligent metasurface performs a 2d dft in the wave domain for doa estimation," in *ICC 2024 - IEEE International Conference on Communications*, 2024, pp. 3445–3451.

[15] F. Wang, T. Mi, C. Wang, R. Xiong, Z. Wang, and R. C. Qiu, "Source localization and power estimation through riss: Performance analysis and prototype validations," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2025.

[16] J. Zhu, Z. Gu, Q. Ma, L. Dai, and T. J. Cui, "A self-controlled reconfigurable intelligent surface inspired by optical holography," *Nature Electronics*, vol. 8, pp. 1108–1118, Nov. 2025. [Online]. Available: https://doi.org/10.1038/s41928-025-01482-3

[17] S. Li, T. Mao, G. Liu, F. Zhang, R. Liu, M. Hua, Z. Gao, Q. Wu, and G. K. Karagiannidis, "Intelligent metasurface-enabled integrated sensing and communication: Unified framework and key technologies," *IEEE Wireless Communications*, vol. 33, no. 1, pp. 216–223, Feb. 2026.

[18] M. Cao, X. Liu, H. Zhang, Y. C. Eldar, and H. Zhang, "Hybrid near-field and far-field localization with holographic MIMO," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 10 560–10 575, 2026.

[19] Z. Zhang and L. Dai, "Wavenumber-domain signal processing for holographic MIMO: Foundations, methods, and future directions," *IEEE Communications Standards Magazine*, vol. 10, no. 2, pp. 127–133, Jun. 2026.

[20] Y. Shen, J. Zhang, S. Jin, and B. Ai, "Intelligent reflecting surface assisted wireless communication with intensity detection," *IEEE Transactions on Wireless Communications*, vol. 19, no. 12, pp. 8316–8328, Dec. 2020.

[21] J. Huang and et al., "Channel sensing for holographic interference surfaces based on the principle of interferometry," *IEEE Transactions on Wireless Communications*, vol. 23, no. 7, pp. 7953–7966, 2024.

[22] J. M. Kahn and J. R. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265–298, Feb. 1997.

[23] J. Armstrong, "OFDM for optical communications," *Journal of Lightwave Technology*, vol. 27, no. 3, pp. 189–204, Feb. 2009.

[24] J. R. Fienup, "Phase retrieval algorithms: A comparison," *Applied Optics*, vol. 21, no. 15, pp. 2758–2769, Aug. 1982.

[25] Y. Shechtman, Y. C. Eldar, O. Cohen, H. N. Chapman, J. Miao, and M. Segev, "Phase retrieval with application to optical imaging: A contemporary overview," *IEEE Signal Processing Magazine*, vol. 32, no. 3, pp. 87–109, May 2015.

[26] E. J. Candès, T. Strohmer, and V. Voroninski, "PhaseLift: Exact and stable signal recovery from magnitude measurements via convex programming," *Communications on Pure and Applied Mathematics*, vol. 66, no. 8, pp. 1241–1274, Aug. 2013.

[27] E. J. Candès, Y. C. Eldar, T. Strohmer, and V. Voroninski, "Phase retrieval via matrix completion," *SIAM Review*, vol. 57, no. 2, pp. 225–251, 2015.

[28] I. Waldspurger, A. d'Aspremont, and S. Mallat, "Phase recovery, MaxCut and complex semidefinite programming," *Mathematical Programming*, vol. 149, no. 1–2, pp. 47–81, 2015.

[29] P. Netrapalli, P. Jain, and S. Sanghavi, "Phase retrieval using alternating minimization," *IEEE Transactions on Signal Processing*, vol. 63, no. 18, pp. 4814–4826, Sep. 2015.

[30] E. J. Candès, X. Li, and M. Soltanolkotabi, "Phase retrieval via wirtinger flow: Theory and algorithms," *IEEE Transactions on Information Theory*, vol. 61, no. 4, pp. 1985–2007, Apr. 2015.

[31] Y. Liu, "On solving phase retrieval problems via neural networks: A unified framework with provable guarantees," *EURASIP Journal on Advances in Signal Processing*, vol. 2026, no. 1, p. 15, 2026.

[32] R. Balan, P. Casazza, and D. Edidin, "On signal reconstruction without phase," *Applied and Computational Harmonic Analysis*, vol. 20, no. 3, pp. 345–356, May 2006.

[33] A. S. Bandeira, J. Cahill, D. G. Mixon, and A. A. Nelson, "Saving phase: Injectivity and stability for phase retrieval," *Applied and Computational Harmonic Analysis*, vol. 37, no. 1, pp. 106–125, Jul. 2014.

[34] Y. C. Eldar and S. Mendelson, "Phase retrieval: Stability and recovery guarantees," *Applied and Computational Harmonic Analysis*, vol. 36, no. 3, pp. 473–494, May 2014.

[35] A. Conca, D. Edidin, M. Hering, and C. Vinzant, "An algebraic characterization of injectivity in phase retrieval," *Applied and Computational Harmonic Analysis*, vol. 38, no. 2, pp. 346–356, Mar. 2015.

[36] P. Grohs, S. Koppensteiner, and M. Rathmair, "Phase retrieval: Uniqueness and stability," *SIAM Review*, vol. 62, no. 2, pp. 301–350, 2020.

[37] D. Gabor, "A new microscopic principle," *Nature*, vol. 161, no. 4098, pp. 777–778, May 1948.

[38] E. N. Leith and J. Upatnieks, "Reconstructed wavefronts and communication theory," *Journal of the Optical Society of America*, vol. 52, no. 10, pp. 1123–1130, Oct. 1962.

[39] I. Yamaguchi and T. Zhang, "Phase-shifting digital holography," *Optics Letters*, vol. 22, no. 16, pp. 1268–1270, Aug. 1997.

[40] B. Gao, Q. Sun, Y. Wang, and Z. Xu, "Phase retrieval from the magnitudes of affine linear measurements," *Advances in Applied Mathematics*, vol. 93, pp. 121–141, 2018.

[41] P. Walk, H. Becker, and P. Jung, "Ofdm channel estimation via phase retrieval," in *2015 49th Asilomar Conference on Signals, Systems and Computers*. IEEE, 2015, pp. 1161–1168.

[42] G. Zhou, Z. Peng, C. Pan, and R. Schober, "Individual channel estimation for ris-aided communication systems—a general framework," *IEEE Transactions on Wireless Communications*, vol. 23, no. 9, pp. 12 038–12 053, 2024.

[43] A. Abdallah, A. Celik, M. M. Mansour, and A. M. Eltawil, "Multi-agent drl for distributed codebook design in ris-aided cell-free massive mimo networks," *IEEE Transactions on Communications*, vol. 73, no. 5, pp. 3283–3297, 2025.

[44] Z. Yu, J. An, L. Gan, H. Li, and S. Chatzinotas, "Weighted codebook scheme for ris-assisted point-to-point mimo communications," *IEEE Wireless Communications Letters*, vol. 14, no. 5, pp. 1571–1575, 2025.